

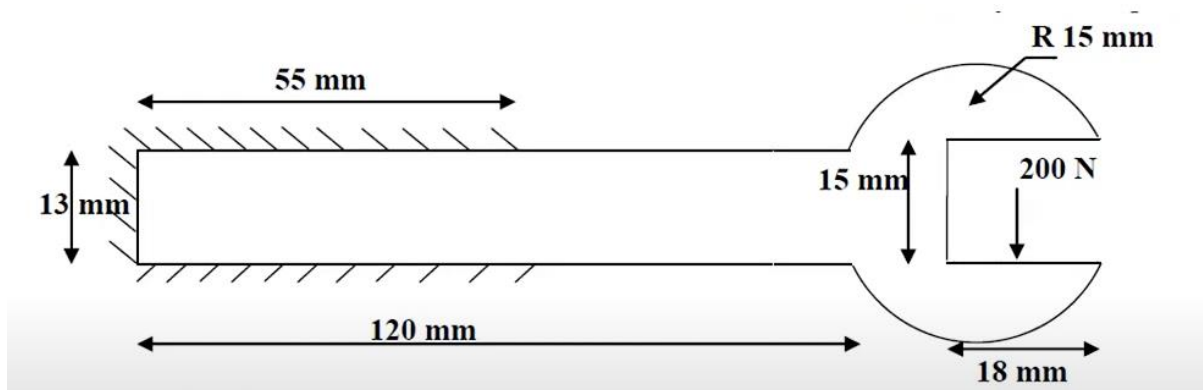
## Practical 7

### Title – Analysis of Machine Components using 3D Elements

#### Problem Statement

In the Spanner under plane stress, find deformed shape and determine the maximum stress distribution.  $E=2 \times 10^5 \text{ N/mm}^2$ ,  $t=3\text{mm}$ , Poisson's ratio = 0.27,

Analysis assumption – plane stress with thickness is used.



#### Procedure

##### Step 1 Preferences

Open Ansys APDL – Preferences – Structural – OK

##### Step 2 Pre-processor

Pre Processor → Add/Edit/Delete → Add → Solid → 8 node 183 → Ok →

Options → Element Behaviour → Plane strsw/thk

##### Step 3 Real constant

Add → Ok → Thickness → 3 → Ok → Close

##### Step 4 Material

Material properties → Material Models → Structural → Linear → Elastic →

Isotropic → Add  $E=2e5$  →  $\nu=0.27$  → OK → Close

### Step 5 Modeling

Modelling → create → Areas → rectangle → By 2 Corneres → X=0, Y=0, Width=120, Height=13 → Ok

Modelling → create → Areas → Solid Circle → X=133, Y=6.5, Radius=15 → Ok

Operate → Booleans → Add → Areas → Pick Rectangle → Pick Circle → Ok

Modelling → create → Areas → rectangle → By 2 Corneres → X=133, Y=0, Width = 15, Height=13 → Ok

Operate → Booleans → Subtract → Areas → Pick Rectangle → Pick Rectangle → Ok

WorkPlane → Offset WP by Increments → X, Y, Z offsets → 55,0,0 → Degrees → XY,YZ,ZX → 0,0,90 → ok.

### Step 6 Meshing

Meshing → Mesh Tool → Mesh Areas → Quad → Free → Mesh → pick all → ok

Mesh Tool → Refine → pick all → Level of refinement → 3 → ok.

### Step 7 Loads

Loads → Define loads → apply → Structural → Displacement → on Areas → select the Spanner end → apply → DOFs to be constrained → ALL DOF → ok.

**Loads → Define loads – apply – Structural – Force/Moment → on Nodes  
→ select node → apply → direction of Force → FY → Force/Moment value (–  
200) → ok.**

### **Step 8 Solution**

**Solution – Solve – Current LS - OK**

### **Step 8 General Results**

**Plot Results → Deformed Shape → def+ undeformed → ok.**

**Plot results → contour plot → Element solu → Stress → Von Mises Stress → ok  
(the stress distribution diagram will be displayed).**